Electrical series circuits

So far we've been looking at applications of first order differential equations, but we do also want to look at least at some applications of second order equations.

So we'll now turn our attention toward electrical circuits. Depending on the components included in the circuit, we'll sometimes model these with first order differential equations, and sometimes with second order differential equations. We'll look at both scenarios.

LRC series circuits

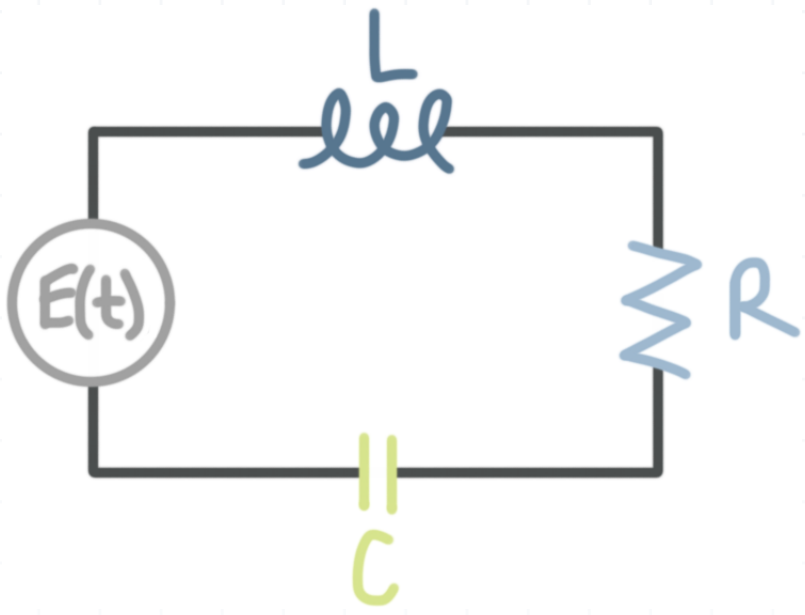
Electrical series circuits are another real-world system that we can model with differential equations. We'll usually see some collection of these four components in a series circuit:

- The inductance L, measured as a constant in henries h
- The resistance R, measured as a constant in ohms Ω
- The capacitance C, measured as a constant in farads f
- The impressed voltage $E(t)$

We sketch these pieces in a circuit that looks something like this:

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Kirchhoff's second law

In a circuit like the one we sketched out above, **Kirchhoff's second law** tells us that the impressed voltage $E(t)$ on a closed loop must be equal to the sum of the voltage drops in the loop.

In other words, if we apply a voltage to the circuit, like plugging in a battery, we're impressing a certain voltage $E(t)$ onto the system. Doing so creates a current $i(t)$ that moves around the circuit.

The voltage drop that occurs when the current encounters the inductor is $L(di/dt)$, the voltage drop that occurs when the current encounters the resistor is Ri, and the voltage drop that occurs when the current encounters the capacitor is $q(t)/C$, where $q(t)$ is the charge on the capacitor at any time t. The current $i(t)$ is related to the capacitor's charge $q(t)$ by $i = dq/dt$.

If we use this last equation $i = dq/dt$ to write everything in terms of q, we can write Kirchhoff's second law as

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$$L\frac{di}{dt} + Ri + \frac{1}{C}q = E(t)$$

$$L\frac{d}{dt}\left(\frac{dq}{dt}\right) + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

When $E(t) = 0$, the electrical vibrations in the circuit are **free**; they're **forced** when $E(t) \neq 0$. And if the circuit is missing one of these components, Kirchhoff's second law still holds. For instance, for a circuit without an inductor, the impressed voltage is still equivalent to the voltage drops over the resistor and capacitor.

$$R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

Or for a circuit without a capacitor, the impressed voltage is still equivalent to the voltage drops over the inductor and resistor.

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} = E(t)$$

With some understanding now of series circuits, let's do an example where we solve one of these differential equations.

Example

Find the current i after connecting a 9-volt battery to a series circuit with inductance 1.5 h and resistance $5\ \Omega$.

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There's no capacitor in the system, which means we can simplify the equation given by Kirchhoff's second law by eliminating the capacitor term.

$$L\frac{di}{dt} + Ri = E(t)$$

Substituting $L = 1.5 = 3/2$, $R = 5$, and $E(t) = 9$, we get

$$\frac{3}{2}\left(\frac{di}{dt}\right) + 5i = 9$$

$$\frac{di}{dt} + \frac{10}{3}i = 6$$

This is now a linear differential equation in standard form, so we'll find the integrating factor.

$$I(t) = e^{\int \frac{10}{3}\, dt}$$

$$I(t) = e^{\frac{10}{3}t}$$

Multiplying through the differential equation by the integrating factor, we get

$$\frac{di}{dt}e^{\frac{10}{3}t} + \frac{10}{3}ie^{\frac{10}{3}t} = 6e^{\frac{10}{3}t}$$

$$\frac{d}{dt}(ie^{\frac{10}{3}t}) = 6e^{\frac{10}{3}t}$$

Integrate both sides, then solve for the current i.

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$$\int \frac{d}{dt}(ie^{\frac{10}{3}t})\,dt = \int 6e^{\frac{10}{3}t}\,dt$$

$$ie^{\frac{10}{3}t} = \frac{9}{5}e^{\frac{10}{3}t} + C$$

$$i = \frac{9}{5} + Ce^{-\frac{10}{3}t}$$

In other words, this solution equation models the current i in the circuit at any time t.

It's particularly interesting to realize that the second term in the equation, $Ce^{-\frac{10}{3}t}$, approaches 0 as $t \to \infty$. For that reason, we call this second term the **transient term**. Any terms that remain after the transient terms drop away are the **steady-state part** of the solution. In this case, $9/5$ is the steady-state current.

The second order circuit equation

In the last example we solved a first order linear differential equation, but when the inductor, resistor, and capacitor are all present in the circuit, Kirchhoff's second law gave us

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$Lq'' + Rq' + \frac{1}{C}q = E(t)$$

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This is a second order linear differential equation. The associated homogeneous equation is

$$Lr^2 + Rr + \frac{1}{C} = 0$$

This is a quadratic equation, so using the quadratic formula the roots of the equation are

$$r = \frac{-R \pm \sqrt{R^2 - \frac{4L}{C}}}{2L}$$

So we can say that the circuit is

- **overdamped** if the discriminant is positive, $R^2 - 4L/C > 0$, in which case we'll find distinct real roots,
- **critically damped** if the discriminant is zero, $R^2 - 4L/C = 0$, in which case we'll find equal real roots, and
- **underdamped** if the discriminant is negative, $R^2 - 4L/C < 0$, in which case we'll find complex conjugate roots.

Let's look at an example with the second order equation.

Example

Given $L = 0.5$ henry (h), $R = 4.5$ ohms (Ω), $C = 0.1$ farad (f), $E(t) = 0$, $q(0) = q_0$ coulombs (C), and $i(0) = 0$, find the charge $q(t)$ on the capacitor in a circuit with an inductor, resistor, and capacitor.

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Plugging what we know into the second order equation given by Kirchhoff's second law, we get

$$L\frac{d^2q}{dt^2} + R\frac{dq}{dt} + \frac{1}{C}q = E(t)$$

$$0.5q'' + 4.5q' + \frac{1}{0.1}q = 0$$

$$q'' + 9q' + 20q = 0$$

The associated characteristic equation for this second order homogeneous equation is

$$r^2 + 9r + 20 = 0$$

$$(r+5)(r+4) = 0$$

$$r = -5, \ -4$$

Because we found distinct real roots, we know the circuit is overdamped. We also know that it's overdamped because the discriminant is positive.

$$R^2 - 4L/C > 0$$

$$4.5^2 - 4(0.5)/0.1 > 0$$

$$20.25 - 20 > 0$$

$$0.25 > 0$$

With distinct real roots, we can say that the general solution and its derivative are given by

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$$q(t) = c_1 e^{-5t} + c_2 e^{-4t}$$

$$q'(t) = i(t) = -5c_1 e^{-5t} - 4c_2 e^{-4t}$$

Substituting the initial conditions $q(0) = q_0$ and $i(0) = 0$ into these equations, we get

$$q_0 = c_1 e^{-5(0)} + c_2 e^{-4(0)}$$

$$q_0 = c_1 + c_2$$

and

$$0 = -5c_1 e^{-5(0)} - 4c_2 e^{-4(0)}$$

$$0 = -5c_1 - 4c_2$$

$$c_2 = -\frac{5}{4}c_1$$

Substituting this value into the first equation gives

$$q_0 = c_1 - \frac{5}{4}c_1$$

$$q_0 = -\frac{1}{4}c_1$$

$$c_1 = -4q_0$$

Plugging this back into the equation for c_2, we get

$$c_2 = -\frac{5}{4}(-4q_0)$$

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$$c_2 = 5q_0$$

So the general solution can be written as

$$q(t) = -4q_0e^{-5t} + 5q_0e^{-4t}$$

$$q(t) = 5q_0e^{-4t} - 4q_0e^{-5t}$$

$$q(t) = q_0e^{-4t}(5 - 4e^{-t})$$

This solution equation models the charge q on the capacitor at any time t.

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